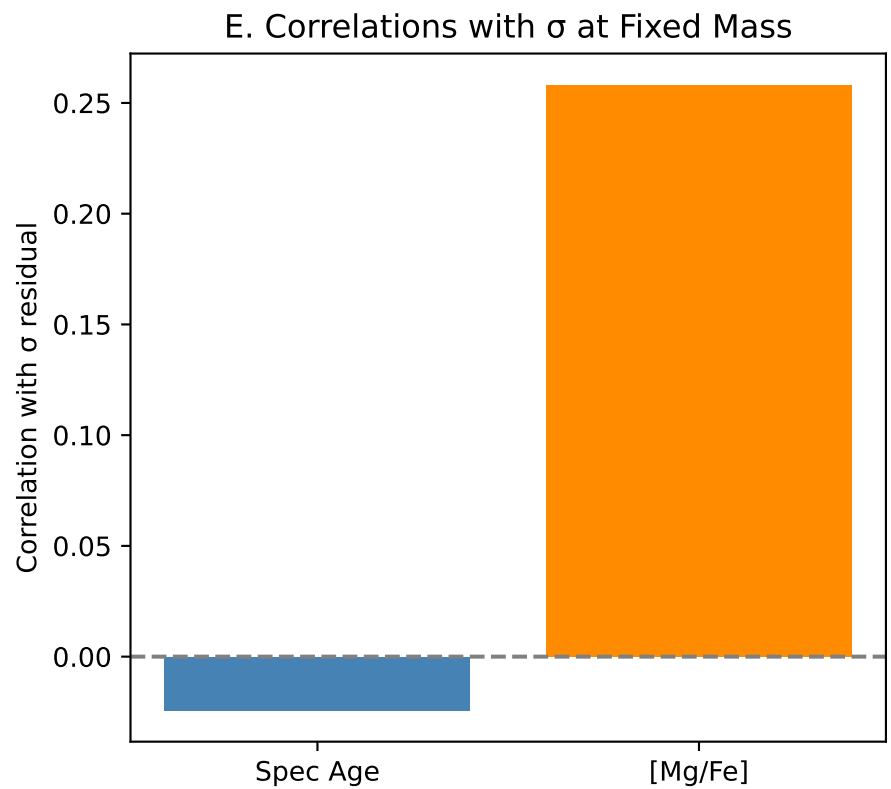
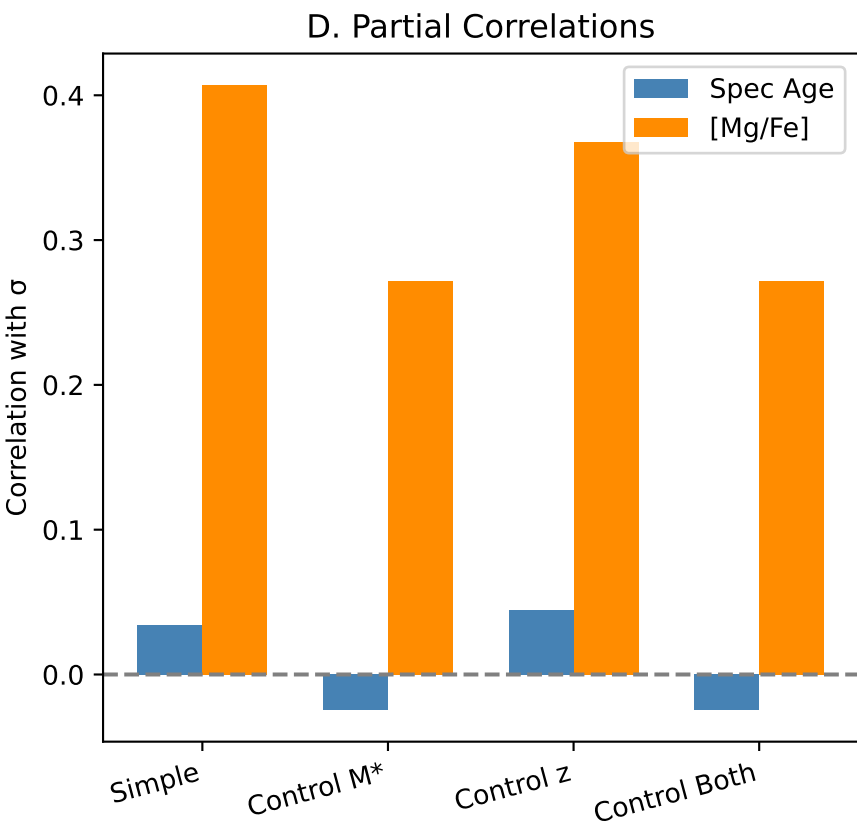
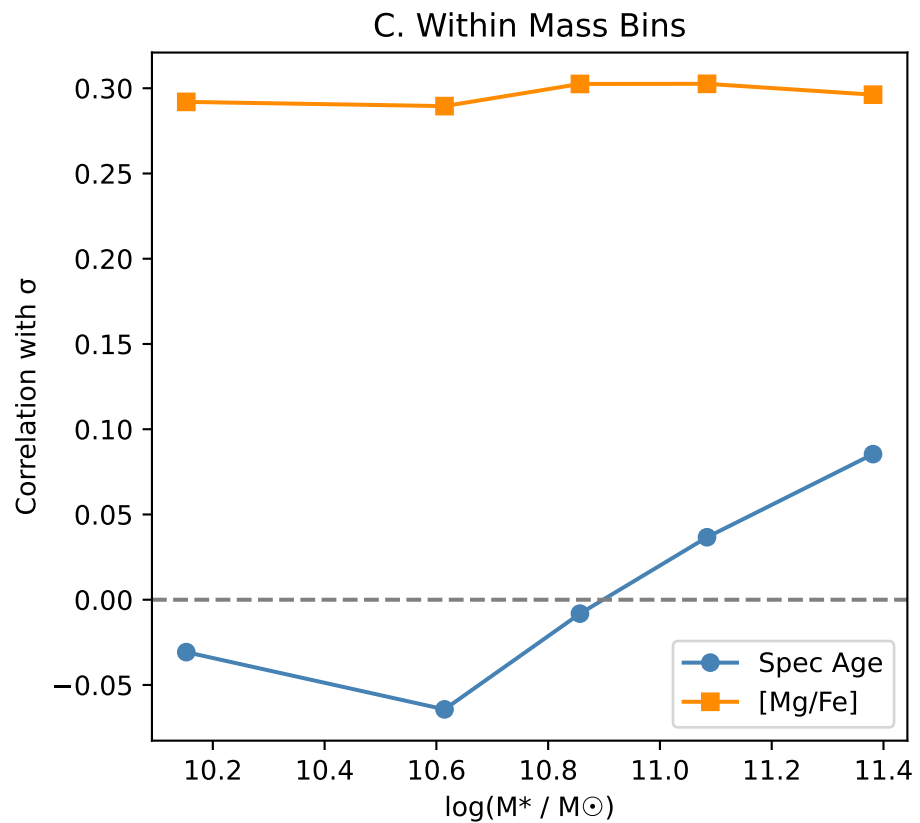
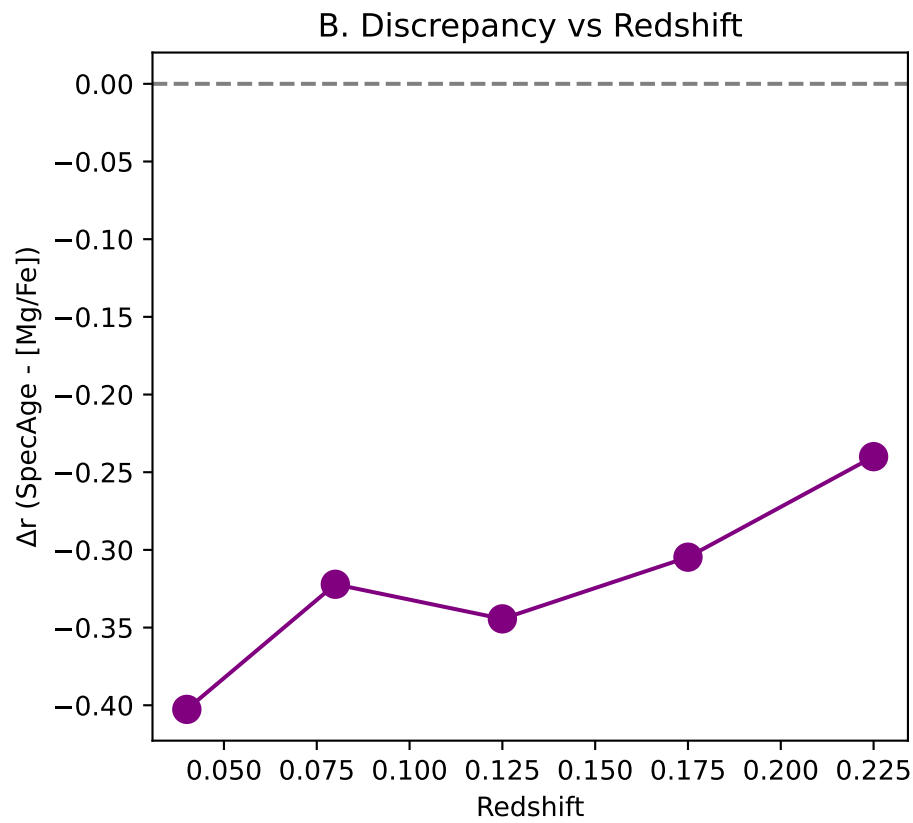
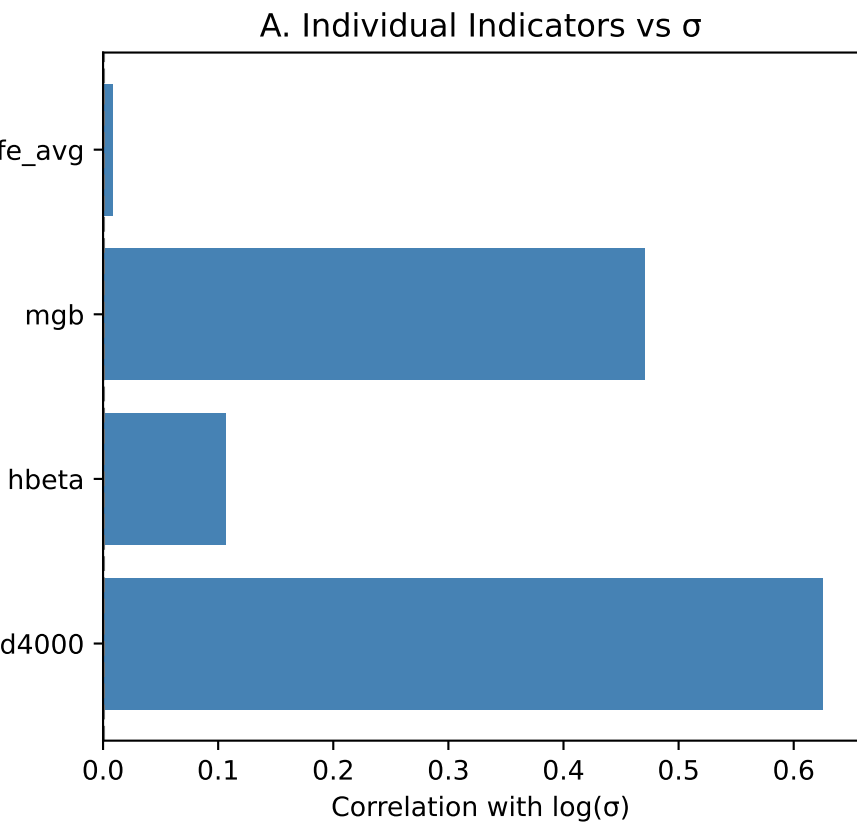




CONTROL TEST SUMMARY

Sample: 361,532 galaxies
Redshift: 0.02 - 0.25

KEY FINDINGS:

1. D4000 and H β have OPPOSITE correlations with σ , explaining weak composite signal.
2. Discrepancy is STABLE across redshift (not a selection effect).
3. [Mg/Fe]- σ relation weakens at fixed mass but spectroscopic age remains uncorrelated.
4. After controlling for M^* and z :
 $\Delta r = -0.297$
5. At fixed mass (σ residual):
 $r(\text{SpecAge}, \sigma_{\text{resid}}) = -0.024$
 $r([\text{Mg/Fe}], \sigma_{\text{resid}}) = 0.258$

INTERPRETATION:

Spectroscopic ages are systematically DECOUPLED from gravitational potential, while nucleosynthesis ratios are not. This pattern is consistent with TEP.